Demand function $P = \frac{R}{x}$

$$P = 35 + \frac{7}{2}x - x^2.$$

Example 3.17

A firm has the marginal revenue function given by $MR = \frac{a}{(x+b)^2} - c$ where x is the output and a, b, c are constants. Show that the demand function is given by $x = \frac{a}{b(p+c)} - b$.

Solution:

$$\text{Given MR} = a(x+b)^{-2} - c$$

$$R = \int a(x+b)^{-2}\,dx - c\int dx$$

$$R = \frac{a(x+b)^{-1}}{-1} - cx + k$$

$$R = -\frac{a}{x+b} - cx + k$$

When $x = 0$, $R = 0$

$$\therefore\ 0 = -\frac{a}{b} - c(0) + k$$

$$k = \frac{a}{b}$$

$$R = -\frac{a}{x+b} - cx + \frac{a}{b}$$

$$= \frac{-ab + a(x+b)}{b(x+b)} - cx$$

$$R = \frac{ax}{b(x+b)} - cx$$

Demand function $P = \frac{R}{x}$

$$P = \frac{a}{b(x+b)} - c$$

$$P + c = \frac{a}{b(x+b)}$$

$$b(x+b) = \frac{a}{P+c}$$

$$x = \frac{a}{b(P+c)} - b.$$

To find the Maximum Profit if Marginal Revenue and Marginal cost function are given:

If 'P' denotes the profit function, then

$$\frac{dP}{dx} = \frac{d}{dx}(R - C) = \frac{dR}{dx} - \frac{dC}{dx} = MR - MC$$

Integrating both sides with respect to x gives ,

$$P = \int (MR - MC)\,dx + k$$

Where k is the constant of integration. However if we are given additional information, such as fixed cost or loss at zero level of output, we can determine the constant k. Once P is known, it can be maximum by using the concept of maxima and minima.

Example 3.18

The marginal cost $C'(x)$ and marginal revenue $R'(x)$ are given by $C'(x) = 50 + \frac{x}{50}$ *and* $R'(x) = 60$. The fixed cost is ₹200. Determine the maximum profit.

Solution:

$$\text{Given C}(x) = \int C'(x)\,dx + k_1$$

$$= \int \left(50 + \frac{x}{50}\right) dx + k_1$$

$$\text{C}(x) = 50x + \frac{x^2}{100} + k_1$$

When quantity produced is zero, then the fixed cost is 200.

i.e. When $x = 0$, $c = 200$

$$\Rightarrow k_1 = 200$$

Cost function is C(x) $= 50x + \frac{x^2}{100} + 200$ (1)

The Revenue $R'(x) = 60$

$$R(x) = \int R'(x)\,dx + k_2$$

$$= \int 60\,dx + k_2$$

$$= 60x + k_2$$

When no product is sold, revenue = 0

i.e. When $x = 0, R = 0.$

Revenue $R(x) = 60x$ (2)

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